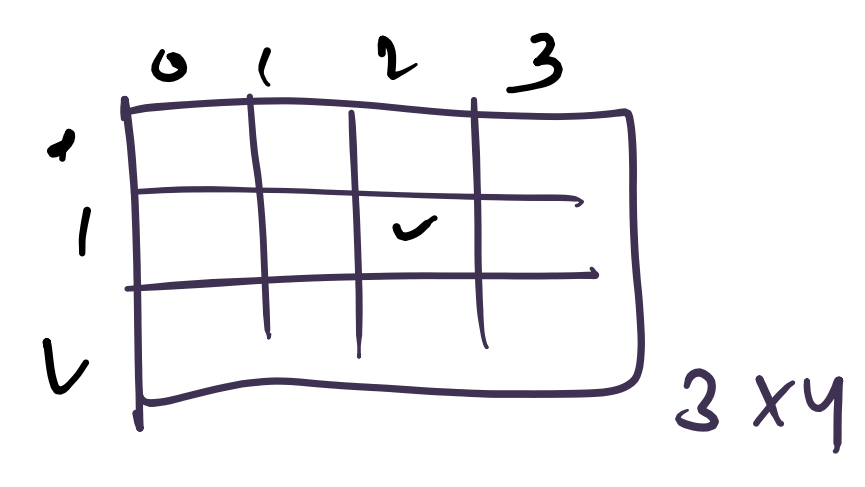
Lec-9

16/09/26 2:16 PM

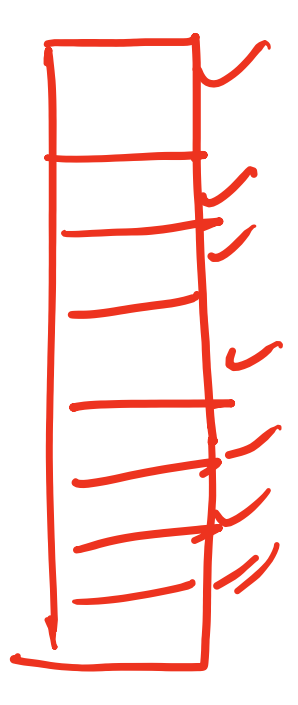
2D Array

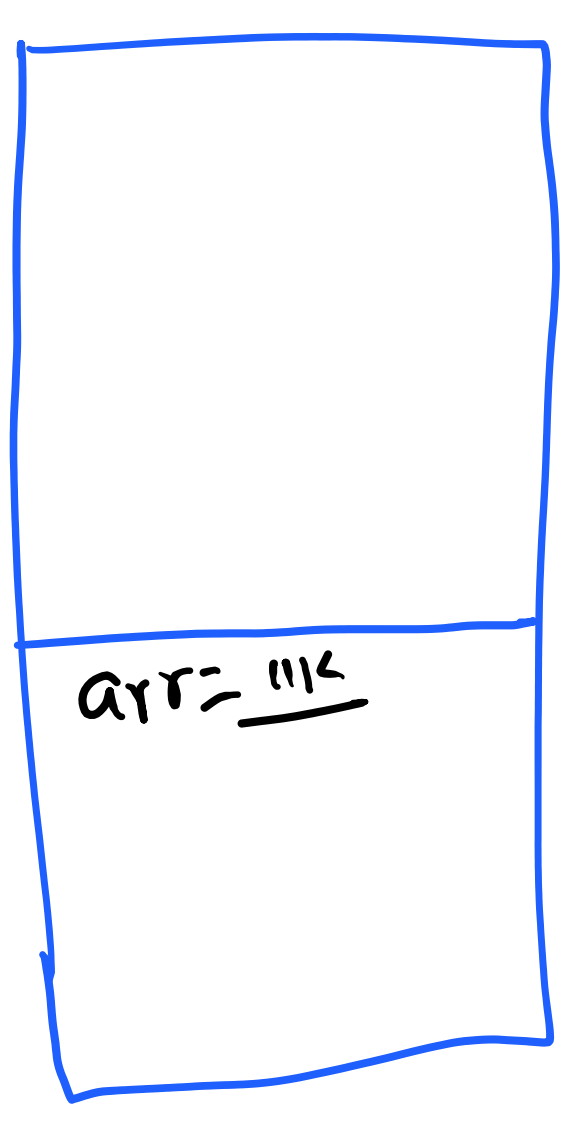


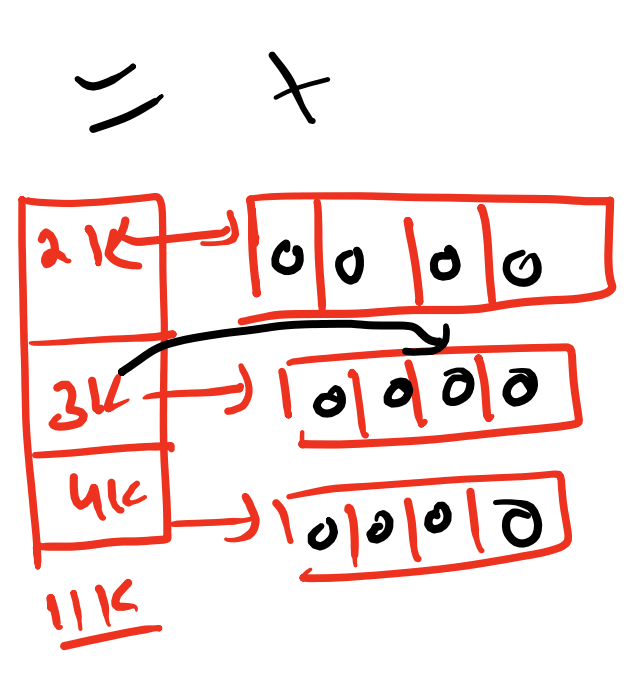
```
public static void main(String[] args) {
    int[][] arr = new int[3][4];
    System.out.println(arr);
}
```

int [ ] [ ] arr = new int [N][M]

1+N



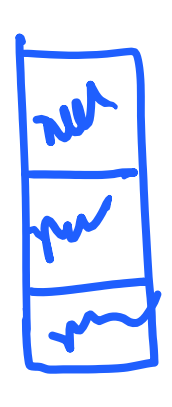


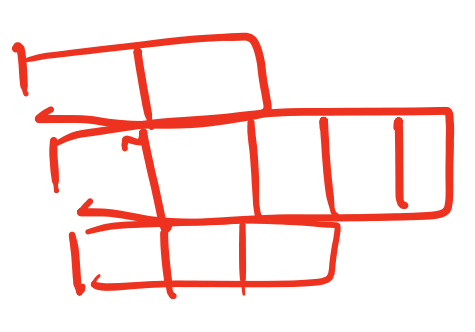


arr[0] → 2k

arr[1][2]

```
int[][] arr = new int[3][];
```

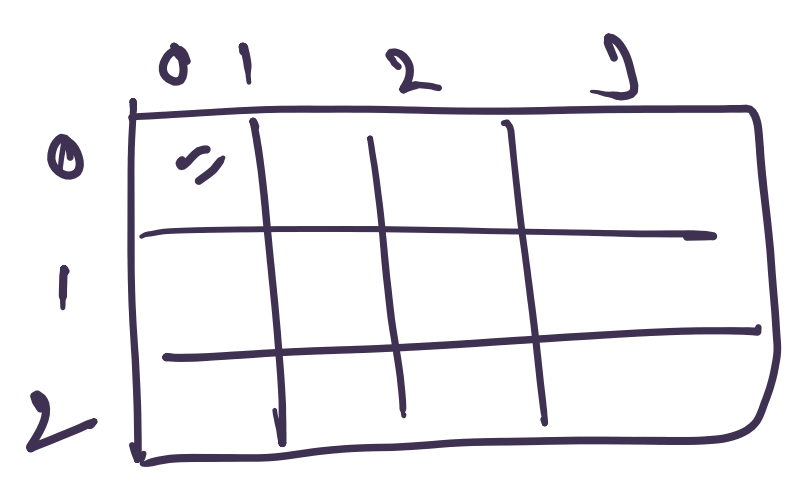




arr[0]= new int[2]

arr[1]= new int[5]

arr[2]= new int[3]



i j

0-0, 0-1, 0-2, 0-3

1-0, 1-1, 1-2, 1-3

2-0, 2-1, 2-2, 2-3

	0	1	2	3	4
0	1	4	7	11	15
1	2	5	8	12	19
2	3	6	9	16	22
3	10	13	14	17	24
4	18	21	23	26	30

1 2 3 10 18 21 13 6 5 4

7 8 9 14 23 26 17 16 12 11

15 19 22 24 30

0-0 1-0 2-0 3-0 4-0

4-1 3-1 2-1 1-1 0-1

0-2 1-2 2-2 3-2 4-2

4-3 3-3 2-3 1-3 0-3

0-4 1-4 2-4 3-4 4-4

	0	1	2	3
0	1	5	9	13
1	2	6	10	14
2	3	7	11	15
3	4	8	12	16

→

1	5	9	13
2	6	10	14
3	7	11	15
4	8	12	16

0-1 → [1,0]

0-2 → [2,0]

0-3 → [3,0]

1-2 → 2-1

1-3 → 3-1

2-3 → 3-2

1	4	7	11	15
2	5	8	12	19
3	6	9	16	22
10	13	14	17	24
18	21	23	26	30

item=13

T.C N log N

→ Row = 0, Col = arr[0].len-1

while ( ) {

if(arr[Row][Col]==item){

return true ✓

else if(arr[Row][Col]>item){

Col--

}else{

Row++

}

	minc		maxc	maxc
1	4	7	11	15
2	5	8	12	19
3	6	9	16	22
10	13	~~14~~	17	24
18	21	23	26	30

minR →

maxR →

1 4 7 11 15 19 22 24 30 26 23 21 18

10 3 2 5 8 12 16 17 14 13 6 9

```
public static void Print(int[][] arr) {
    int minr = 0, maxr = arr.length - 1, minc = 0, maxc = arr[0].length - 1;
    int te = arr.length * arr[0].length; 12
    int c = 0;
    while (c < te) {
        for (int i = minc; i <= maxc; i++) {
            System.out.print(arr[minr][i] + " ");
            c++;
        }
        minr++;
        for (int i = minr; i <= maxr; i++) {
            System.out.print(arr[i][maxc] + " ");
            c++;
        }
        maxc--;
        for (int i = maxc; i >= minc; i--) {
            System.out.print(arr[maxr][i] + " ");
            c++;
        }
        maxr--;
        for (int i = maxr; i >= minr; i--) {
            System.out.print(arr[i][minc] + " ");
            c++;
        }
        minc++;
    }

}
```

1	2	3	4
5	6	7	8
9	10	11	12

minR = ~~0~~ ~~1~~ 2

maxR = ~~2~~ 1 0

minc = ~~0~~ 1 2

maxc = ~~3~~ ~~2~~ 1

C = 4+2+3+1 = 10+2 = 12 → 13

1 2 3 4 8 12 11 10 9 5 6 7 8

minr=1 maxR=~~3~~ 4

minc=~~∞~~ 0 maxc=~~3~~ 4

```
public static int Minimum_Area(int[][] arr) {
    int minr=Integer.MAX_VALUE; +∞
    int minc=Integer.MAX_VALUE; +∞
    int maxr=Integer.MIN_VALUE; −∞
    int maxc=Integer.MIN_VALUE; −∞
    for (int i = 0; i < arr.length; i++) {
        for (int j = 0; j < arr[0].length; j++) {
            minr=Math.min(minr, i);
            minc=Math.min(minc, j);
            maxr=Math.max(maxr, i);
            maxc=Math.max(maxc, j);
        }
    }
}
```

1 4

	0	1	2	3	4
0	0	0	0	0	0
1	0	1	0	0	1
2	0	0	0	0	0
3	1	0	0	0	0
4	0	0	1		0

4-1+1 = 4

(4-0+1) = 5